LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Co-axial and counter rotating propellers, forces acting on propeller
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Co-axial and counter rotating propellers, forces acting on propeller. Constant Speed Unit (CSU)
5. Specific Objectives	12.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT diagrams; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Name two types of propellers based on rotation direction.
(c) Define the Constant Speed Unit (CSU).

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Co-axial propeller configuration • Counter rotating propeller system • Advantages of counter rotation • Forces acting on propeller blades • Blade element momentum theory	L & Ds	PPT, IB	15 min
(b)	• Constant Speed Unit (CSU) operation • Governor mechanism function • Pitch change mechanisms • Hydraulic vs mechanical control • CSU response to throttle changes	L & Ds	PPT, PDF	20 min
(c)	• Propeller efficiency factors • Torque effect mitigation • Vibration causes and solutions • CSU maintenance basics • Operational limitations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how counter rotation reduces torque effect.
(b) Describe the role of the CSU in maintaining constant RPM.
(c) List three forces acting on a propeller blade.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational advantages of co-axial propellers versus conventional single propellers, referencing LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Effect of propeller rotation on aircraft and methods used to counteract it	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min